



HK IMO Training 2023–2024
Problem Set 14
For 20 January 2024



Problem 53. In a square $ABCD$, let E be a point on the diagonal AC . Let H be the orthocentre of $\triangle ADE$, and let M and N be the midpoints of AD and CD respectively. Prove that $HM \perp EN$.

Problem 54. Find all positive integers n with the following property: For any real numbers x and y , there exist real numbers $a_1, a_2, \dots, a_n$ such that

$$x = a_1 + a_2 + \dots + a_n \quad \text{and} \quad y = \frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}.$$

Problem 55. Initially, there are two numbers r and $\frac{1}{r}$ written on the blackboard where r is a positive rational number. Every time we can choose two numbers on the blackboard and write their arithmetic mean or their harmonic mean on the blackboard. Find all r such that the number 1 can be written on the blackboard in finitely many steps.

Problem 56. There is a calculator which displays a certain integer $a > 1$ initially. The calculator has three buttons. By pressing these three buttons respectively, we can change the current displayed number n to $d(n)$, $\sigma(n)$ and $\varphi(n)$ respectively. Prove that we can make the calculator display any integer $b > 1$ after pressing the buttons finitely many times. (Here, $d(n)$ is the number of positive divisors of n , $\sigma(n)$ is the sum of all positive divisors of n , and $\varphi(n)$ is the number of integers from 1 to n which are relatively prime to n .)